STA302 Research Proposal Handout

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1. Introduction

The goal of this handout is to help you think about how to develop a research proposal.

Broadly, a research proposal is a document that communicates a proposed research plan to a particular audience. A strong research proposal makes a well-supported **argument**: that the research question is worth investigating and that your proposed approach is a reasonable way to investigate it.

One useful way to think about this is to put yourself in the position of your reader. Suppose you were responsible for deciding which research projects should be allocated time, attention, or other resources. What would you like to know? You might ask:

- What is the researcher trying to understand?
- Why does this research matter? Who benefits from the research?
- What is already known about the topic? What would this project add to what we know?
- Are the proposed data appropriate for the question? Does the researcher have a sensible plan for analyzing those data?
- What ethical issues, limitations, or consequences should be considered?

A research proposal should help its reader answer these questions.

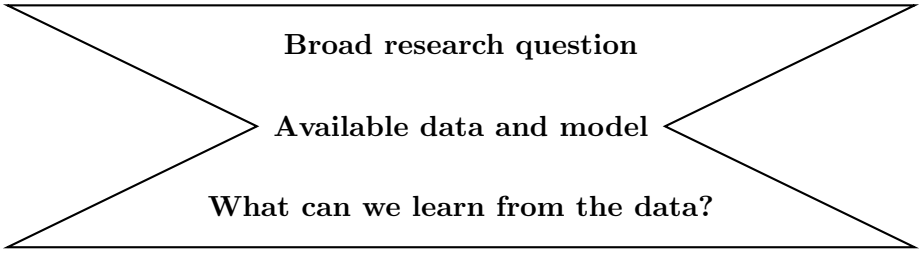
The research proposal you will create in STA302 serves an additional purpose—it allows us to assess whether you have met the relevant learning outcomes for the course. It is therefore important to read the assignment instructions and rubrics carefully before you begin.

2. Developing a Research Question

In the course folder you will find an excellent guide titled "Tips for Developing a Research Question," written by Dory Abelman.

Here I want to emphasize one feature of the process that is especially important for this project: developing a research question is *iterative*. It usually involves moving back and forth between a bird's-eye view of the research problem and the technical nitty-gritty.

You will often begin with a broad topic that interests you. At this stage, the question may still be vague. At this point it can be useful to think of the conceptual space as an *hourglass*:



At the top is the broad question that motivates the research. At the narrowest point are the data and variables available to you. For this assignment, your data must also be suitable for *a linear model*. At the bottom of the hourglass, you return to the broader question: what can your results reasonably tell us, and what can they not? For this assignment, you should be also able to have a rough sense of which aspect of the model you will construct will answer the broader question.

Do not expect to arrive at the final version of your research question immediately. The process is iterative: you may narrow or even change the broad question significantly as you read, inspect your data, and learn more about the topic.

3. Reading and Research

The literature can inform all steps of this research process throughout. It can help you narrow the question, choose meaningful variables, justify your modelling choices, and understand how your results relate to existing work.

When you hold your broad research question in mind and approach articles, you can perform **goal-directed reading** (Leora Freedman from the ELL program has an excellent article on this). For this case, it is useful to approach articles with the assignment goals in mind. Here are what the rubrics say about an satisfactory literature review:

- What each article tells you about the relationship between your response and predictors is explicitly explained and connected back to your research question.
- Your research question addresses a gap in the existing literature or approaches an existing problem in a novel way.

That means you should not begin by reading every article word for word. Read with the purpose of deciding whether and how the article will contribute to these two points.

Start by reading the abstract, introduction, and conclusion. Preview the section headings, figures, and tables. If the paper does not seem promising for either of the goals above, leave it aside and move on. At the early stages it is more about finding the right papers to focus on.

If a paper does seem useful, then read more closely. Ask:

- Who wrote the paper, and where was it published?
- Who seems to be the intended audience?
- What question is the paper trying to answer?
- What data does it use?
- What method does it use? Does it provide precedent for using linear regression?
- What variables are important?
- What is the central result?
- Why do the authors think the result matters?

You may also find that reading one article suggests more specific searches. That is good. Literature searching and literature reading are also iterative.

It can help to read with either (or both) a **critical eye** and a **collaborative eye**.

- With a critical eye, ask what the limitations of the study are. Are there problems with the methodology? Are there limitations in the data? Are there important relationships between variables that the authors do not consider? Does the study support all of the conclusions the authors draw?
- With a collaborative eye, imagine that you are trying to continue the conversation started by the authors. What questions remain unanswered? What directions for future research do they suggest? Could their methods be applied to a different population, dataset, or question?

Don't worry if you encounter terminology, methods, or arguments that you do not immediately understand. Mark the difficult passage, keep reading, and return to it later if it turns out to be important. Your understanding of one paper often improves after you have read several related papers.

4. Literature review

Once you have done this for several articles, you are ready to begin writing the **literature review**. A literature review is an *opinion-driven* introduction to the relevant research landscape that explains where your own question fits within it.

In other words, your reader should understand why you included each paper. Here are some plausible reasons (non-exhaustive, of course):

- Perhaps previous papers study the same general relationship you are interested in, but use different populations.
- Perhaps previous papers rely on a particular type of data that creates important limitations.
- Perhaps previous papers study related questions, but none addresses the specific relationship you want to investigate.
- Or perhaps a paper uses a method that can be applied to your own problem even though its substantive question is different.

Needless to say, a literature review is therefore **not** a sequence of article summaries. Architecturally, avoid giving each article its own paragraph. Instead, organize the literature around ideas, themes, findings, or limitations. For example:

> The literature has largely focused on A, particularly A1 (Author, 2012) and A2 (Author, 2015). More recent work has extended this approach by considering B (Author, 2020). However, these studies rely primarily on [type of data or population], leaving open the question of whether [your gap or extension].
>
> There is little or no work directly examining [specific gap or extension]. However, related research provides a useful starting point. Studies of [nearby case A] have found [relevant finding] (Author, 2018), while work on [nearby case B] has used [relevant method or variable] to study a similar relationship (Author, 2021). These findings and methods suggest a useful way to investigate [your research question].

Notice what this does. The articles are not simply reported one after another. They are being *used* to build an argument. Ultimately, the literature review should help answer the question: **Why does this research matter given what others have already done?**

5. Data and Ethics Discussion

The purpose of this section is to establish that your dataset is suitable for the project, sufficiently complex to meet the assignment requirements, and appropriate to use.

Begin by briefly describing where the dataset came from, why it was originally collected, and how it was collected. Cite the original source. Then summarize the response and predictor variables using appropriate plots or numerical summaries. Confirm that the dataset satisfies the assignment requirements, justify why the response is suitable for linear regression, and explain your choice of interaction term.

The ethics discussion should not simply label the dataset as "ethical" or "unethical." Instead, make a reasoned argument addressing two questions:

1. **Is this dataset trustworthy?**
2. **Was this dataset collected and made available in an ethically appropriate way?**

Use concepts from the first ethics module to justify your answers. For trustworthiness, among the things to consider include: who produced the data, how the variables were measured, what documentation is available, and whether there are reasons to question the accuracy or completeness of the dataset. Do not simply state that the dataset comes from a reputable source or is publicly available; explain why those facts do or do not support your confidence in the data.

For ethical collection, consider how the data were obtained and shared. Relevant issues may include consent, privacy, potential harms, the population from which the data were collected, and the purpose for which they were originally gathered.

6. The Rest of the Proposal

The remaining sections are relatively straightforward. Still, each part of the proposal should serve the larger goal described above. For the specific requirements, I will refer to the rubric.

Writing a proposal can be genuinely fun. You get to learn more about the world, about statistics, about your teammates, and perhaps even about yourself. Happy writing!